ForecastLabAI — modular monolith architecture

one process · 19 feature modules in app/features/ · one PostgreSQL + pgvector · external LLMs

source: ForecastLabAI @79e8dce · arrows = calls / data flow

React dashboard
Dashboard · Explorer · Visualize · Knowledge · Chat · Control Center

no auth layer —
API keys set at runtime
via /config, never logged

REST /forecasting · /registry · …

WebSocket /agents/stream

FastAPI backend — one process, one deploy

GenAI

streamed replies over WebSocket
actions wait for human approval

agent tools call into:
forecasting · backtesting
registry · scenarios · rag

agents
PydanticAI · approvals

rag
index · retrieve

agents → LLM · rag → embeddings

LLM providers
Anthropic · OpenAI

Ollama (optional, local)

ML — train → test → rank → register

registry
runs · aliases · SHA-256 check

model_selection
rank candidates → champion

backtesting
chronological folds

forecasting
train · predict

jobs / batch — portfolio runs

runs

forecasts

Decision — what-if · why · KPIs

scenarios
baseline vs what-if; only the regression
model re-forecasts, others: heuristic

explainability
explains a forecast run (reads registry)

analytics
KPIs and series straight from sales

```
# app/core/config.py
agent_default_model =
  "anthropic:claude-sonnet-4-5"
agent_fallback_model =
  "openai:gpt-4o"
rag_embedding_model =
  "text-embedding-3-small"
# ollama: nomic-embed-text
```

Data foundation — everything reads it

sales

sales history

upsert

featuresets
time-safe features (no leakage)

data_platform
stores · products · sales_daily

ingest
validate + upsert sales

also in the process: dimensions · seeder (The Forge) · config · ops · demo

SQLAlchemy async · asyncpg

PostgreSQL 16 + pgvector (pgvector/pgvector:pg16)
sales_daily · model_run · deployment_alias · job · document_chunk · agent_session

document_chunk.embedding = Vector(1536) — rag ranks by pgvector cosine_distance